

CollabXR: A Survey of XR Collaboration

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Abstract—A systematic review of 557 papers detailing collaboration research utilizing augmented, virtual, and mixed reality (AR/VR/MR) across various publication venues spanning 2000 to 2023 is presented. The aim is to evaluate the approaches used in Extended Reality (XR) collaboration and explore future opportunities within this domain. A taxonomy has been developed to provide a structured understanding of the current research landscape. Based on this taxonomy, publications are categorized by input type (e.g., 360 video, point cloud), device type (e.g., HMDs, HHDs), interaction modality, collaboration mode (e.g., one-to-one, one-to-many), and evaluation type (e.g., Presence and Immersion). This classification helps in systematically analyzing the diverse approaches and technologies employed in XR collaboration research. Further analysis of 395 papers from the top ten publication venues provides insights into current trends in publication types (e.g., System Design, User Study) and application areas (e.g., Immersive Learning, Healthcare). Key findings from the review highlighted the importance of low-latency, real-time communication in XR collaboration, which was essential for inducing a sense of presence and immersion among users and enabling natural interactions like interacting with and manipulating virtual objects. Supporting multiple users, beyond traditional one-on-one interactions, with asymmetric compatibility ensured diverse teams could engage effectively within the XR collaboration space. In terms of collaboration styles and user experience principles, the focus was on spatial, remote, and asynchronous interaction, enhancing user experience through presence, intuitive interaction design, and considerations for comfort and ergonomics. The main gaps and opportunities in this research area have been identified by examining current trends in XR collaboration. These trends underscore the need for adopting multi-user platforms, integrating AI, and developing industry-specific applications. However, challenges such as network optimization, privacy concerns, and the refinement of interaction models persist. Addressing these issues is crucial for advancing the capabilities and usability of XR collaboration tools.

Index Terms—Virtual Reality, Augmented Reality, Remote Collaboration

I. INTRODUCTION

THE landscape of remote collaboration has undergone significant transformation, particularly driven by the COVID-19 pandemic, and has emerged as a prominent technique in the field of computer graphics [1]. Remote collaboration offers a wide range of benefits, tools, and capabilities that enable teams to work together in real-time effectively, regardless of geographical location. Technological advancements and the widespread adoption of Extended Reality (XR) technologies, encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), are transforming remote collaboration by creating immersive virtual environments for meetings, training, and design work [2], [3].

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The lack of comprehensive guidelines and methodologies for designing and evaluating XR collaboration system presents a notable challenge, especially in establishing structured pipelines. This pipeline is particularly evident when it comes to evaluating the usability and performance of XR solutions, as identifying suitable evaluation strategies, variables to examine, and appropriate data collection methods becomes inherently complex, making it difficult to address specific research questions in creating XR collaboration experiences that meet both user needs and performance standards.

Prior surveys have investigated such guidelines and methodologies, mostly focusing on particular areas such as remote assistance and training [4], education [5], security and privacy [6], and cultural heritage [7], typically drawing from very focused data sources. However, specifically focus on guidelines and methodologies for designing, developing, and evaluating XR collaboration solutions that includes AR, VR, and MR technologies needed. None of the aforementioned approaches specifically focus on guidelines and methodologies for designing, developing, and evaluating XR collaboration solutions that includes AR, VR, and MR technologies.

The objective of this paper is to provide a comprehensive overview of XR collaboration by analyzing a large volume of publications sourced from diverse data repositories. Through this analysis, the aim is to elucidate a structured pipeline for the design, development, and evaluation of XR collaboration system. By synthesizing insights from a broad spectrum of research, the survey endeavors to provide actionable guidance for practitioners and researchers alike, offering a roadmap to navigate the complexities inherent in XR collaboration. Additionally, the survey sheds light on the implications for future research and practice in this domain, facilitating the forecasting of future trends and charting the research trajectory for XR collaboration. Through these efforts, there is an aspiration to contribute to the advancement of XR technologies and their application in fostering collaborative environments across various sectors.

The contributions of this survey are as follows:

- A **comprehensive taxonomy** of methods and approaches for XR collaboration.
- A **comparative analysis** of XR collaboration system in terms of features, compatibility, scalability, user interface, and communication tools.
- The **identification and discussion** of challenges and future trends, including technical constraints, user comfort, data privacy, and integration with existing workflows.
- An **interactive visualization** tool for exploring the research papers and a publicly available data set of the analysis.

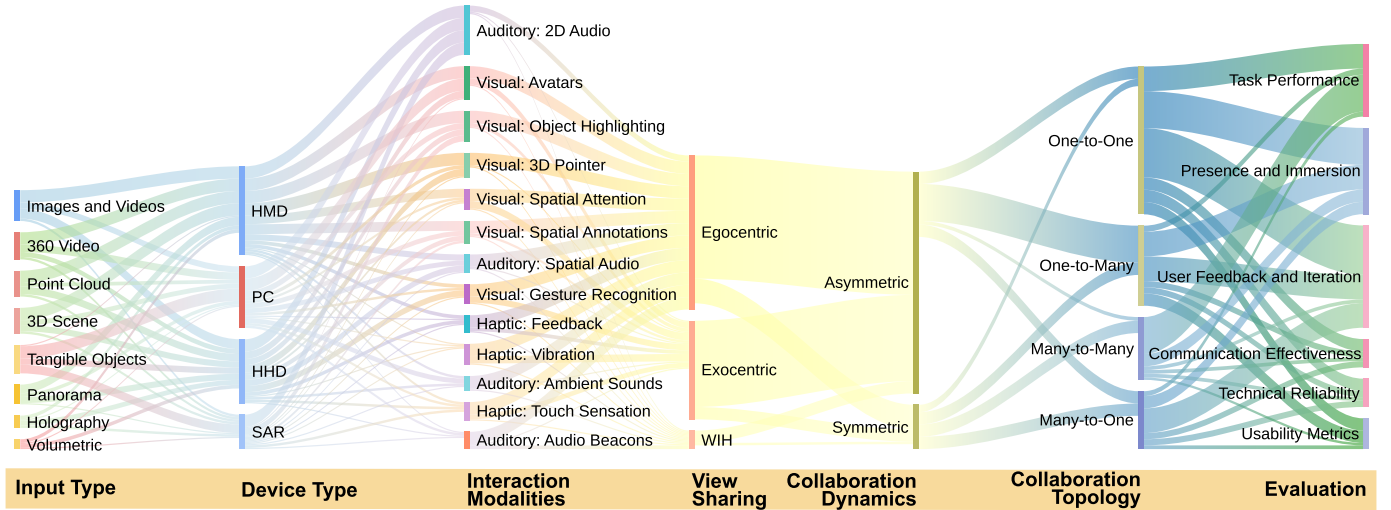


Fig. 1. A total of 557 papers that employed XR collaboration research between 2020 and 2023 were classified by (left-to-right): system input type, device type, interaction modality, communication cues, and evaluation method in user studies.

II. BACKGROUND AND SCOPE

This section outlines the foundational concepts, theories, and technologies related to XR collaboration. It then defines the scope of this survey, specifying aspects that will be covered and excluded.

XR Collaboration. Collaboration involves individuals or groups working together towards a common goal [8]. In Computer-Supported Cooperative Work (CSCW), solutions like *groupware* have been developed for knowledge sharing among distributed collaborators [9]–[11], focusing on the space and time of collaboration [12].

XR collaboration refers to interactions within Extended Reality (XR) environments, including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), where users can communicate and collaborate in immersive digital spaces. Mann et al. [13] align XR with the MR continuum [14], while others, like Al-Adhami et al. [15], view XR as encompassing both MR and VR. Paradiso and Landay [16] introduced ‘*cross reality*’ blending physical and virtual environments through sensor/actuator networks.

Despite varying definitions, XR technologies share a common goal: augmenting reality to create experiences beyond the physical world, enabling richer collaboration through immersive technologies.

Key Concepts in XR Collaboration. Presence is the feeling of ‘*being there*’ within a virtual environment, while co-presence refers to the awareness of sharing that space with others. Effective collaboration in XR relies heavily on achieving high levels of both presence and co-presence. Social interaction theories, including social presence theory and media richness theory, help to elucidate how individuals gauge communication effectiveness in virtual contexts. Social presence is especially significant as it measures the degree to which a medium allows users to feel connected and understood. Moreover, embodied cognition theory highlights the influence of the body on cognitive processes, indicating that XR technologies that replicate physical movement and

spatial awareness can improve collaboration by creating a more instinctive and immersive experience.

Related Survey. Over the years, numerous compelling research studies have explored augmented reality and virtual reality. Various independent reviews in the literature that covered significant aspects of AR, VR, and MR and inspired this survey are listed in Table I.

Wallace et al. [17] reviewed how Computer-Supported Cooperative Work (CSCW) research has changed over time, especially with the emergence of new technologies like smartphones and wearables. They analyzed ACM Conference on CSCW proceedings from 1990 to 2015 to understand these changes. The findings show that CSCW research now emphasizes describing collaborative work environments in practice rather than developing new systems in labs. There’s also a lack of bibliographic work and focus mainly on single-device studies. The study suggests challenges for CSCW research in adapting to technological changes effectively.

Boletsis et al. [18] reviews VR locomotion techniques from 2014 to 2017, identifying 36 relevant articles covering 11 methods such as real-walking and joystick-based locomotion. It highlights a predominant focus on VR technology over user experience since the VR revival, with most techniques relying on physical interaction for navigation. The study introduces a typology categorizing VR locomotion into four types: motion-based, room scale-based, controller-based, and teleportation-based.

A 10-year survey by Dey et al. [19] reviewed 369 AR user studies from 2004–2014. They found an increasing trend of involving handhelds in AR user studies. They also provided more application areas, adding Education and Medical as areas with a high number of publications with user studies. They have confirmed that most user studies conducted in a laboratory setting as 75% were lab-based evaluations.

A comprehensive review of collaborative systems based on AR and MR from 2013–2018 has been proposed by de Belen et al [20]. Their study focuses on how users in

remote asynchronous MR environments can create digital information and re-visit project meetings. The meta-review of collaborative MR systems shows that 103 of 259 systems use a local collaboration of multiple users in a shared physical environment. They also pointed out that the provision of non-verbal cues such as gaze and pointing cues were identified as two important factors in enhancing collaboration quality in MR spaces.

A more recent systematic exploration of the mixed-reality space by Ens et al. [21] confirms that “collaborative MR technology is now mature enough to focus squarely on human needs.” This includes looking at how hand-held AR can be used to look at complex collaboration in space as well as time and symmetry. They pointed to mixed presence as an issue that still needs to be addressed to match the complexity of real-world collaboration. Furthermore, the roles of the users in groups were highlighted as another issue that is yet to be addressed in groups.

Merino et al. [22] reviewed 458 papers on evaluations in mixed and augmented reality (MR/AR) from 2009 to 2019. It categorized them by type, research topic, evaluation scenario, cognitive aspects, and context. Two main groups were identified: technology-centric and human-centric studies. Of the 458 papers, 248 involved user studies with 5,761 participants, predominantly in laboratory settings. This survey primarily centers on the influence of technological innovation on collaborative XR research, while incorporating traditional CSCW concepts. Relevant papers have been gathered from major conference proceedings, including CHI, ISMAR, IEEE VR, VRST, and CSCW.

Survey Scope. This survey specifically focus on XR collaboration across VR, AR, and MR, examining the following aspects: A review of existing XR tools that facilitate collaborative work, including software applications and hardware systems (*Technologies and Platforms*). An exploration of how users interact within XR environments, emphasizing user interfaces, avatars, and gestures (*Interaction Models*). Exclusions from this survey will include traditional Computer-Supported Cooperative Work (CSCW) approaches that do not integrate XR technologies, as well as single-user experiences that do not involve collaboration.

Expanding the survey to include a structured pipeline for XR collaboration design, development, and evaluation could provide actionable guidance for practitioners and researchers.

III. METHODOLOGY

This study employs a systematic literature review (SLR) approach following the PRISMA (Preferred Reporting Items

for Systematic Reviews and Meta-Analyses) methodology¹. The process is structured to ensure comprehensive and unbiased identification, selection, and analysis of relevant studies. Detailed methods for data collection, coding, and analysis were implemented to facilitate systematic synthesis.

A. Data Collection Method

The data collection process was divided into four key stages: identification, screening, eligibility assessment, and inclusion. A systematic search was conducted across multiple electronic databases, including PubMed, Scopus, IEEE Xplore, and Google Scholar. The search utilized carefully selected keywords and Boolean expressions, focusing on terms associated with AR, VR, MR (e.g., “augmented reality”, “virtual reality”, “mixed reality”, “AR”, “VR”, “MR”) combined with collaboration-related terms. Regular expressions were crafted to maximize the retrieval of relevant studies from diverse sources. A summary of the search strategy is illustrated in Figure 3.

B. Screening and Selection

The initial search results yielded **1,425 articles**, which were imported into reference management software where duplicates were removed. The remaining articles underwent title and abstract screening based on predefined inclusion and exclusion criteria. A total of **1,403 abstracts** were screened for eligibility. Of these, **22 abstracts** were inaccessible due to paywalls or other restrictions. Articles that failed to meet the inclusion criteria were excluded. Following this rigorous screening process, the final dataset comprised studies meeting all eligibility requirements.

C. Inclusion and Exclusion Criteria

Data extraction was carried out using a standardized form to ensure consistency and thoroughness. The form captured key information, including study objectives, methodologies, collaboration styles, and evaluation techniques. This systematic approach facilitated accurate categorization and analysis of the selected studies. To ensure the relevance and quality of the included studies, the following criteria were applied:

Inclusion Criteria:

- Publications in peer-reviewed conferences or journals.
- Articles written in English.
- Studies focusing on XR collaboration involving empirical data or experimental results.
- Publications from the years 2000 to 2023.

Exclusion Criteria:

- Grey literature, such as conference abstracts, theses, and dissertations.
- Articles unavailable in full text.
- Publications not related to XR collaboration (e.g., articles using search terms for unrelated purposes).
- Duplicate publications (e.g., when identical results were reported in both conference proceedings and journal articles, the more comprehensive version was retained).

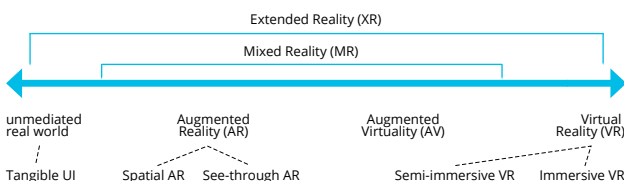
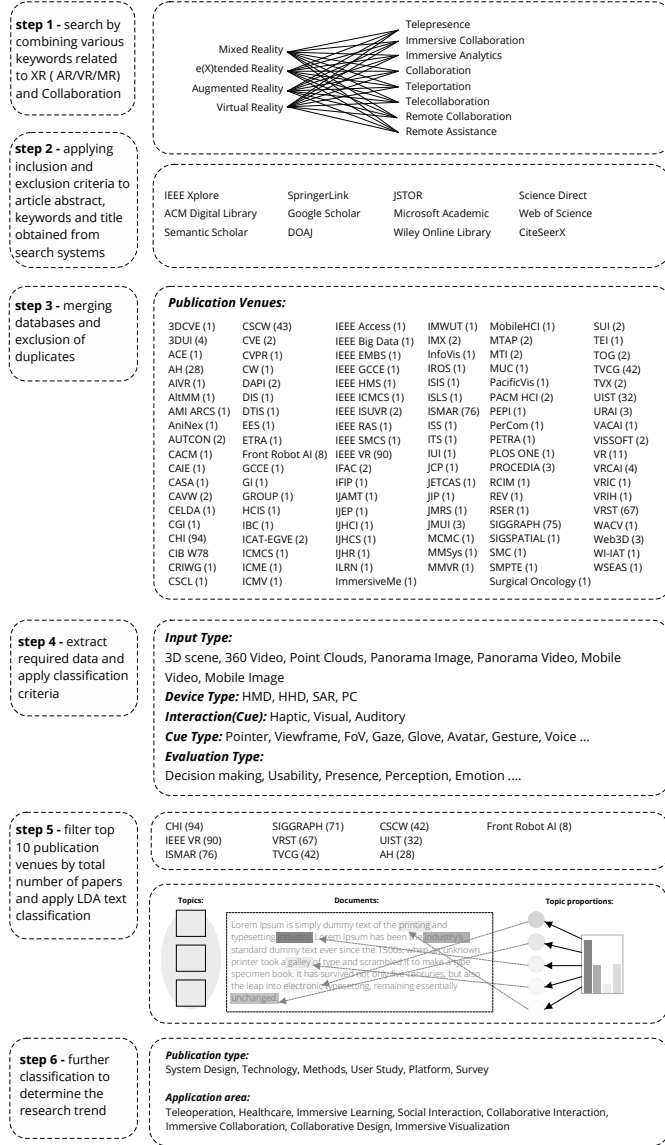


Fig. 2. Reality-Virtuality Continuum adapted from Milgram & Kishino [14].

¹<https://www.prisma-statement.org/prisma-2020-statement>

TABLE I
SUMMARY OF EXISTING SURVEY REVIEW PAPERS INVOLVING COLLABORATION IN AR/VR/MR.

Review Paper	Year	Time Period	Summary	Test Conditions
				# of Papers
Wallace et al. [17]	2017	1990–2015	analyzed publications across four classification schemes	1209
Boletsis et al. [18]	2017	2014–2017	identified four distinct VR locomotion types	36
Dey et al. [19]	2018	2005–2014	identified two areas with high publications	291
de Belen et al. [20]	2019	2013–2018	overviews on collaboration in MR	259
Ens et al. [21]	2019	1995–2018	focuses on the design possibilities of collaborative AR	111
Merino et al. [22]	2020	2009–2019	provided guidance for future evaluations of MR/AR approaches	458
This review	2024	2000–2023	analyzed publications focus on collaborative XR	557



IV. CLASSIFICATION AND CODING

To systematically categorize and analyze the studies included in this review, we developed a structured approach encompassing both the creation of a taxonomy and the application of a coding scheme.

A. Taxonomy

A taxonomy was developed to systematically classify and organize the key concepts and findings from the included studies (see Figure 1). The taxonomy was designed to cover various aspects relevant to the research topic, including:

1) **Input Type:** Refers to the method or medium through which users interact with and input data into the system. This includes the ways in which data is captured, represented, and input into mixed reality systems to create immersive experiences.

a) **Images and Videos:** Most of the earlier collaboration systems were developed using 2D images and video. These systems typically utilized cameras connected to common mobile, desktop, or embedded devices [23]–[26]. One example is Zoom, where a collaborator employs a camera's video feed as input and shares it with a remote user in real time [26], [27].

b) **Panorama:** Panoramas, created by stitching images together to provide a wide-angle view, do not necessarily offer a full 360-degree field of view (see Figure 5a). These panoramic views enhance interactivity and control over the viewing experience, allowing for seamless exploration of immersive environments [28], [29].

c) **360-degree Video:** 360-degree videos provide full panoramic views, enhancing immersion and allowing users to explore and interact with environments using 3 degrees of freedom (3DoF). Due to the limited field of view of most webcams, which restricts the capture of remote environments on telepresence platforms, 360-degree video-based telepresence platforms are becoming increasingly popular [30], [31].

d) **Point Cloud:** Point clouds are collections of data points sampled from real-world surfaces, typically generated by 3D scanners or LiDAR, capturing detailed information about physical objects or spaces (see Figure 5b). They enable real-time mapping of environments, allowing remote collaborators to visualize and interact with a shared physical space [32]–[34].

e) **3D Scene:** Virtual worlds serve as environments where users can teleport to simulate interactions with participants through communication and collaboration. These environments are often pre-created using digital graphics software

Fig. 3. The workflow for the data collection and analysis process.

- Non-primary studies such as review articles, commentaries, posters, and demos.
- After applying these criteria, **557** studies were included in the final analysis. This rigorous selection ensured a high-quality dataset for subsequent coding and synthesis.

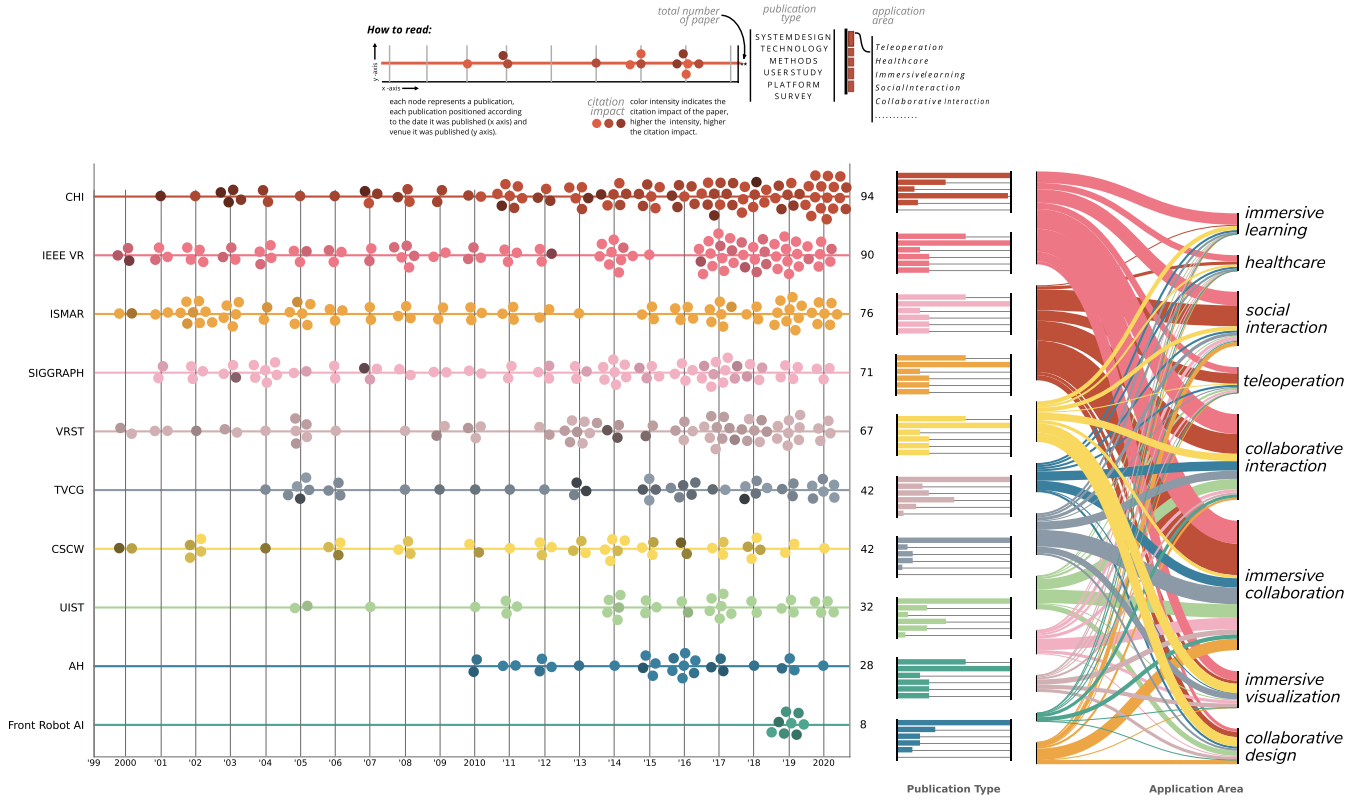


Fig. 4. Distribution of 395 papers that were further assessed to determine the current XR collaboration trend are organized by year of publication, publication venues, citation impact, type of publication, and area of application. An interactive version of this classification is available at <https://zaps.one/collabxr>

or 3D reconstruction software [35]–[38]. Consequently, numerous studies have been conducted within this setup, which is well-suited for immersive learning and collaborative design integration to enhance the sense of presence [39], [40].

f) Holography: This involves interacting with holographic representations or displays, where users manipulate virtual objects that appear as holograms within a mixed reality environment. For example, Holobox is a multi-user holographic display system that enables participants to interact with holographic projections in real time [41]. It is designed for collaborative settings where multiple users can view and manipulate the same 3D data simultaneously.

g) Volumetric: Representations that involve 3D models or volumetric capture, often used to create detailed, interactive digital environments or objects that users can manipulate (see Figure 5c). Users can manipulate and interact with volumetric representations using gestures, controllers, or other input devices within immersive environments.

h) Tangible Objects: This involves using physical objects or props as input devices within mixed reality settings. Tangible objects can be tracked and integrated into virtual environments, allowing users to interact with digital content or control virtual elements through physical manipulation of tangible objects [42].

2) Device Type: Refers to the hardware setup or display environment used to deliver mixed reality experiences (e.g., HMDs, projection-based systems).

a) Desktop or PC: Desktops or PCs allow users to both actively and passively experience MR content [43]. However, these setups typically provide limited immersion.

b) Immersive Rooms/Caves: Immersive rooms or CAVEs are physical spaces equipped with multiple displays or projection surfaces that surround users, creating a fully immersive environment. Users can interact with virtual content or simulations within these spaces using specialized input devices, such as motion trackers or handheld controllers [44].

c) Projection-Based: Projection-Based Systems project XR content onto physical surfaces, such as walls or screens, creating immersive environments for collaborative interactions and presentations. Users can gather around the large screen display to view and engage with XR content simultaneously, making it suitable for group collaboration and shared experiences [45]. Spatial Augmented Reality (SAR) is one example of Projection-Based Systems, where virtual objects are seamlessly integrated into the physical environment through projection, enhancing users' perception of reality and enabling interactive experiences [46].

d) Hand-held Devices (HHD): Handheld devices, such as smartphones and tablets, equipped with cameras, sensors, and powerful processors, can deliver MR experiences. These devices utilize cameras to capture the real world and then overlay virtual elements on the device's screen. Users can interact with these virtual objects through touch gestures, offering a more accessible entry point into MR. This technology has been employed as a collaboration tool in various systems [24],

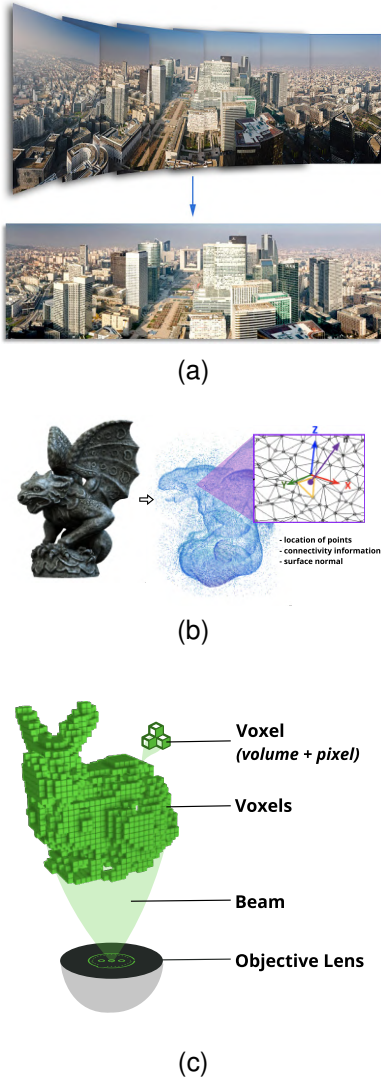


Fig. 5. (a) Stitching a Panorama using multiple images, (b) Point cloud sampled from the real surface, (c) Holograms, also called volumetric displays, are composed of voxels (volume + pixel).

[47]–[49].

e) Head-mounted Display (HMD): HMDs immerse users in digital environments or overlay digital information on the real world. Three types of different head-mounted display technologies exist to superimpose graphics onto the user's view of the real world:

- *Video See-Through (VST) Displays* VST displays use video cameras to capture the real-world environment and overlay digital content onto the user's field of view, eg: Microsoft HoloLens, Magic Leap.
- *Optical See-Through (OST) Displays* OST displays use optical components, such as transparent displays or mirrors, to superimpose digital information onto the user's natural field of view e.g., Google Glass.
- *Fully Immersive or Opaque Displays* block the user's view of the real world and present a completely virtual environment e.g., Oculus Rift, HTC Vive.

3) *Interaction Modalities:* Interaction cues play a vital role in shaping how users engage with the XR environment and collaborate with others. These cues provide guidance and feedback to users, enabling effective communication and collaboration within the mixed reality space. This encompasses visual cues, auditory cues, haptic feedback, and other forms of interaction (see Figure 6).

a) Visual Cues: Visual cues, such as highlighting certain objects or elements within the XR environment to draw attention or indicate significance, also assist users in manipulating virtual objects by providing visual feedback on their position, orientation, and manipulation axis.

- *Avatars* are virtual representations of users, which can range from simple icons to full-body representations or highly detailed, customizable digital versions of the users. They enable users to interact with each other and with the environment, enhancing the sense of presence and social interaction.
- *Object highlighting* serves as a visual cue to draw attention to specific elements within a scene. This technique enhances the user's ability to focus on relevant objects, facilitating better understanding and interaction within virtual environments.
- *Spatial annotations* also known as “see-what-I-see” capability, allow users to draw or highlight objects in real space, providing a way to visually communicate and share insights within a shared environment. This feature enhances collaboration by making virtual notes or marks that are contextually tied to the physical space.
- *3D pointer* used in virtual environments to indicate positions or objects within the space, assisting users in pointing out details or directing attention.
- *Gesture recognition* involves the use of sensors and software to interpret and respond to the user's hand or body movements, allowing for intuitive interaction within mixed reality environments.
- *Spatial attention* ability of a system to detect and respond to where a user is focusing their attention within a 3D

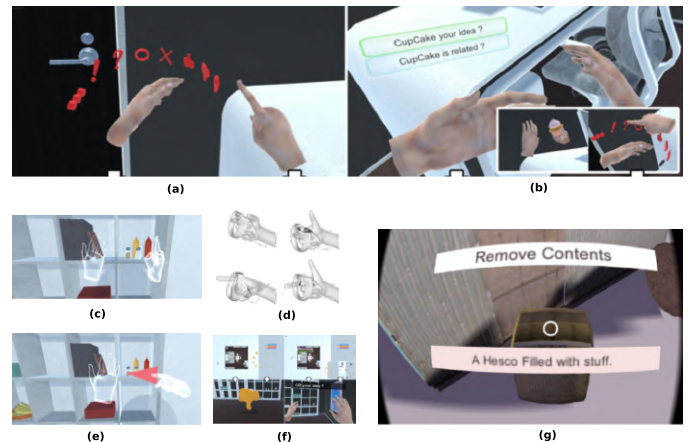


Fig. 6. Interaction techniques in XR collaboration: Drawing annotation (a), viewpoint conversion (b), pointing (c), hand gestures (d), ray to objects (e), overlay icons and label text on top shared objects (f), gaze (g). Diagram adopted from [50].

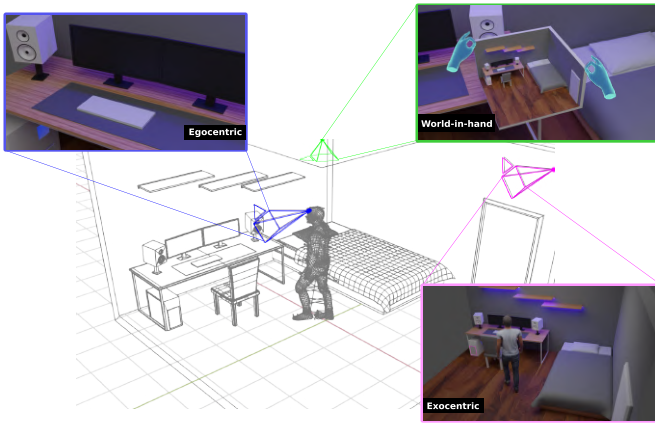


Fig. 7. Visualization of Ego, Exo, and World-in-Hand viewpoints.

space. This can involve tracking eye movements, head orientation, or pointing gestures to determine what the user is looking at or interacting with.

b) Auditory Cues: Auditory cues including ambient sounds, feedback sounds, or notification sounds that provide auditory feedback.

- *2D Audio* refers to traditional stereo or mono audio without spatial positioning.
- *Spatial Audio* is positioned within the virtual environment to create a sense of direction, distance, and location relative to the user's position.
- *Ambient Sounds* enhance the immersive experience by simulating real-world auditory cues, such as nature sounds or city ambiance.
- *Audio Beacons* utilize distinctive audio signals or markers to guide users toward specific locations or objects through auditory cues.

c) Haptic Cues: Haptic cues provide tactile feedback to users, enhancing the sense of immersion and realism in virtual environments. These cues simulate the feel of objects, textures, or forces, allowing users to interact more naturally with digital content.

- *Vibration* simulates physical sensations or notifications through tactile feedback provided by input devices, such as controllers or wearables.
- *Touch Sensation* simulates textures, surfaces, or interactions with virtual objects to enhance the sense of touch within the virtual environment.
- *Feedback (force, Kinesthetic)* simulates forces, resistance, or physical interactions (e.g., pushing, pulling) experienced by users interacting with virtual objects, enhancing realism and immersion.

4) View Sharing: Viewpoint sharing in an MRC environment allows participants to view each other's perspectives, align actions, provide immediate feedback, and interact with digital content in ways that enhance the collaborative experience (Figure 7).

a) Egocentric: This perspective represents the user's point of view, often centered around the user's head and hands. It provides a first-person view, giving users a sense

of being within the environment and interacting directly with the objects around them.

b) Exocentric: This perspective offers an external viewpoint, often from a third-person perspective. It provides an overview of the scene, showing the user's interactions with the environment from outside the user's immediate viewpoint, which is useful for navigation and understanding spatial relationships.

c) World-in-Hand: This perspective enables users to manipulate and interact with a miniature view of the virtual world. It resembles a "bird's-eye view" or "god view," providing users with an overview of the environment. This allows them to control or alter aspects of the collaboration space, facilitating tasks such as positioning, organizing, and editing within the virtual environment.

5) Collaboration Scenarios: XR collaboration setups can vary depending on the combination of devices, tools, and the type of collaboration style supported by the platform. A "collaboration place" refers to a local space, which could be an office, a computer-generated scene, a construction site, or even a space station. The location from which a user joins the collaboration place is termed a remote space. *Participants role* in the collaboration place is defined as follows:

- *Local user:* This refers to a participant who is physically present in the collaboration space where the interaction primarily occurs. This could be someone using VR/AR equipment in a specific physical location, such as an office, construction site, or designated virtual meeting space.
- *Remote user:* This refers to a participant who joins the collaboration from a different location, typically using XR devices or tools to connect virtually to the shared environment. Remote users interact with the local user space and other participants through digital means, enabling virtual presence and participation despite physical distance.

The interaction of participants within MRC environments, based on their roles, responsibilities, and capabilities, significantly dictates the *Collaboration dynamics* of XR applications. Whether participants have equal or varying levels of involvement influences these dynamics, which can be characterized by:

- *Asymmetric* In asymmetric collaboration, the participants have different roles, responsibilities, or capabilities. One party might have more power, knowledge, or resources than the other. For example, in a mentorship relationship, the mentor typically has more experience and knowledge than the mentee.
- *Symmetric* collaboration occurs when all participants have equal roles, responsibilities, and capabilities. Each party contributes to the collaboration in a similar manner. For instance, in a team project where every member has an equal say in decision-making and contributes equally to the work, it's a symmetric collaboration.

Once roles and dynamics are assigned, the way individuals engage in collaborative activities defines the *Collaboration mode*. This term categorizes and defines different approaches

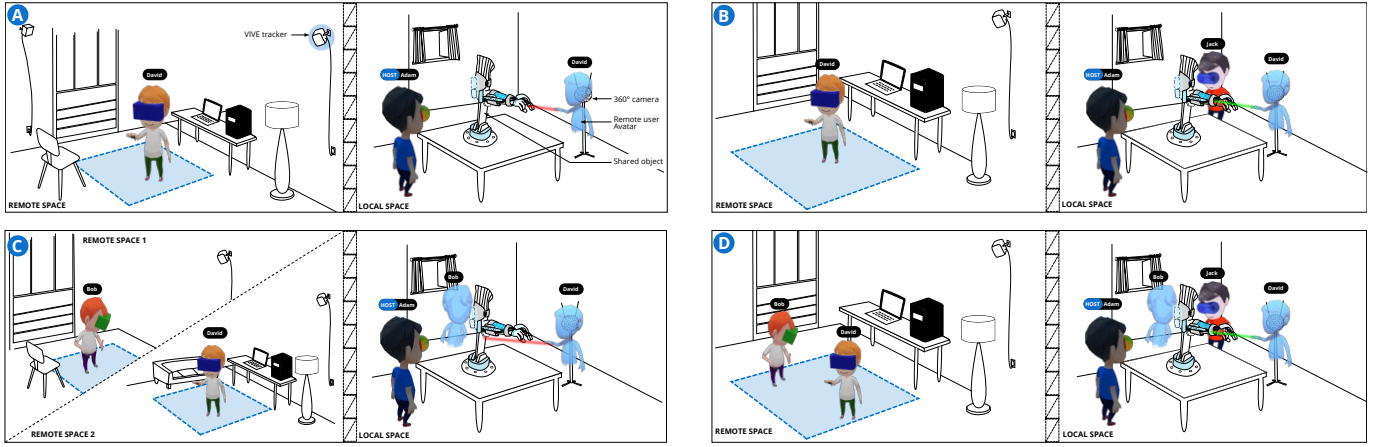


Fig. 8. Different types of collaboration mode in XR collaboration. Remote space to Local space (A) One-to-One ($1 \nleftrightarrow 1$), (B) One-to-Many ($1 \nleftrightarrow n$), (C) Many-to-One ($n \nleftrightarrow 1$), (D) Many-to-Many ($n \nleftrightarrow n$).

to working together, whether in real-time or over extended periods. Examples of collaboration modes include:

- *Asynchronous* collaboration doesn't require all participants to be present or active at the same time. Instead, individuals can work independently on their own schedules and then share their contributions with others. Email, discussion boards, and shared documents are common tools used for asynchronous collaboration.
- *Synchronous* collaboration involves real-time interaction between participants. All parties are active and engaged simultaneously, allowing for immediate feedback and communication. Video conferencing, instant messaging, and live document editing are examples of synchronous collaboration tools.

The communication and interaction between collaborators, influenced by their roles, the chosen collaboration mode, and the overall dynamics of the interaction, require the establishment of a specific *Communication topology*.

- *One-to-One* each collaborator interacts directly with another single collaborator, for example, describes a typical scenario for a meeting between two people, involving bi-directional communication.
- *One-to-Many* one collaborator communicates with multiple other collaborators simultaneously, for instance, a single remote user interacts with multiple hosts at a single location.
- *Many-to-One* multiple collaborators communicate collectively or individually with one central collaborator. For example, a team of engineers collaborates on designing a complex system in XR. They each contribute their ideas and modifications to a shared virtual prototype, which is overseen and reviewed by a project manager.
- *Many-to-Many* multiple collaborators (remote users and local users) interact freely with one another without a central communication hub. For example, architects, interior designers, and clients participate in a virtual design review meeting in XR. They explore and discuss different aspects of a building design together, providing real-time feedback and making collaborative decisions.

6) *Evaluation*: After the development of each MR application, evaluation of the implemented system to understand its potential shortcomings is important. This evaluation is conducted from several perspectives, e.g., based on a user survey, a comparison with a common system, or based on an environmental reference. Evaluation based on user surveys shows the user satisfaction level with the system output. A comparison of results with a common system helps to determine the potential advantages or disadvantages of the MR application in practice, and evaluating the system output by observing environmental references is a practical MR-based navigation application.

a) *Presence and Immersion*: Evaluate the sense of presence and immersion experienced by participants in the XR environment. Evaluation includes measures such as the Igroup Presence Questionnaire (IPQ) and the Networked Minds Measure (NMM).

b) *Usability Metrics*: Assess the overall usability, user satisfaction, and engagement with the XR system. This is measured through standardized questionnaires such as the System Usability Scale (SUS), the User Experience Questionnaire-Short (UEQ-S), and the Usability Metric for User Experience (UMUX-Lite).

c) *Task Performance*: Measure the efficiency and effectiveness of task completion within the mixed reality environment. This includes both subjective and objective measures such as completion time, the NASA Task Load Index (NASA-TLX) or RTLX, and error rate analysis.

d) *Communication Effectiveness*: Evaluate the quality and efficiency of communication between participants in the mixed reality setting. This can be assessed by analyzing the clarity and fidelity of audio and visual communication, conducting conversation analysis (e.g., turn-taking, clarity of speech), and performing observational studies.

e) *Technical Reliability*: Assess the reliability, scalability, and flexibility of the mixed reality platform and associated technologies. This involves monitoring frame rate and latency, as well as generating heatmaps from user tracking data to understand usage patterns and system performance.

f) *User Feedback and Iteration*: Gather qualitative feedback from participants to inform iterative improvements of the system. Methods include conducting qualitative interviews, organizing focus groups, using think-aloud protocols during tasks, and administering post-task surveys to gather user perspectives and suggestions for enhancements.

7) *Empirical Derivation of Topics*: To identify the prominent topics discussed in a substantial number of publications, the Latent Dirichlet Allocation (LDA) [51] method is employed. LDA automatically interprets and clusters words and documents into various topics. This text-mining technique is commonly used in recommendation systems, such as web search engines and advertising applications. LDA is a generative probabilistic model for a corpus, where it treats documents d as random mixtures over latent topics (t), $p(t|d)$, where every topic is characterized by a distribution over words (w), $p(w|t)$. The method uses the following formula:

$$CI = \frac{\text{total number of citations}}{\text{number of years since publication}} \quad (1)$$

Here, $p(w|d)$ represents the probability that a certain word will appear in a particular document on a certain topic. Word-topic distribution $p(w|t)$ and topic-document distribution $p(t|d)$ are initially randomly chosen, and the LDA iteratively updates and estimates these probabilities until the system converges. This process establishes the relationships between documents, topics, and words. All 395 abstracts were input into the LDA model, and a range of parameters were tested. Through empirical analysis, it was determined that six topics best encapsulate publication types, and eight topics suit applications in the field (see Table II and Table III).

To increase the likelihood of capturing prominent and influential research trends and applications within the field of XR collaboration, the papers were organized by their publication venue, and selected those from the top 10 venues. These top venues often feature papers that are considered influential, providing valuable insights into emerging trends, prevalent applications, and innovative approaches in the field. (see Table II and Table III). Through empirical analysis, the following topics were identified.

a) *Publication Type*: In the empirical analysis of topics within XR collaboration, publications are categorized based on their types (see Table II). Below are the primary publication types identified through this analysis:

b) *Application Area*: The specific domains or fields where XR collaboration technologies are being applied or studied were derived through analysis of application areas. These domains represent diverse contexts where XR collaboration solutions are being explored, developed, and deployed to address various challenges and opportunities (see Table III). Following application areas were identified:

B. Coding Scheme

The coding process involved systematically applying the developed taxonomy to the selected studies. This included the following steps:

Initial Training: Coders were trained on the taxonomy and coding scheme to ensure consistency in application. Pilot

TABLE II
SUMMARY OF THE PUBLICATION TYPE.

Publication type	Summary
User Study	<i>user evaluation, user study design</i>
System Design	<i>proof-of-concept, prototype</i>
Platform	<i>system architecture, system integration and scalability security and privacy</i>
Collaborative Interfaces	<i>interface design, interaction techniques, usability testing</i>
Rendering Techniques	<i>visual fidelity, frame rate optimization, real-time rendering</i>
Applications	<i>case studies, application development, use case scenarios</i>

TABLE III
SUMMARY OF THE APPLICATION AREAS.

Application areas	Summary
Healthcare	<i>medical training simulations, patient therapy, surgical planning and rehearsal</i>
Immersive Learning	<i>simulation-based training, virtual classrooms, educational games</i>
Collaborative Design	<i>team-based design processes, interactive prototyping, design reviews</i>
Collaborative Interaction	<i>multi-user simulations, interactive shared environments, co-located interaction</i>
Social Interaction	<i>social platforms, online communities, digital networking</i>
Immersive Collaboration	<i>virtual meetings, remote management, collaborative workspaces</i>
Immersive Visualization	<i>architectural design, scientific visualization, virtual tours</i>
Teleoperation	<i>remote control of machinery, robotic surgery, drone operation</i>

Coding: A subset of papers was coded by multiple coders to test the coding scheme and identify any ambiguities or inconsistencies. Refinement: Based on feedback from the pilot coding, the coding scheme was refined to improve clarity and reliability. Full Coding: Each paper in the review was coded independently by at least two coders. Discrepancies were resolved through discussion and consensus or by involving a third coder if necessary. The coding scheme provided a structured framework for classifying and coding information from the studies. For each study, coders recorded:

Study Identifier: A unique identifier for each study. Input Type: The primary input type used in the study, coded using predefined categories. Device Type: The type of device used in the study, coded using predefined categories. Input Modality: The input modality employed in the study, coded using predefined categories. Additional Variables: Any other relevant

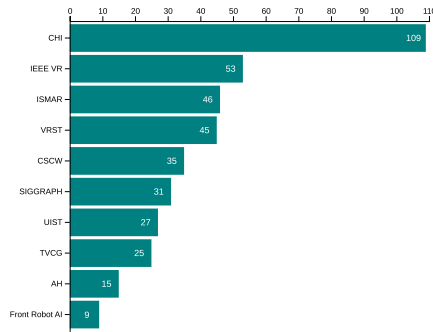


Fig. 9. Distribution of publications across the top 10 venues.

variables identified during the coding process, such as study context, participant demographics, and key findings. To ensure the reliability of the coding process, inter-rater reliability was calculated using Cohen's kappa for categorical variables and intraclass correlation coefficients for continuous variables.

V. RESULTS

The results of the systematic literature review are now reported and organized according to the taxonomy introduced in section IV to answer the research questions, identify gaps in the literature, and discuss opportunities for future research.

A total of 557 papers were analyzed, published across 135 venues. Key publication sources include ACM CHI (109 papers), IEEE VR (53 papers), IEEE ISMAR (46 papers), ACM VRST (45 papers), and ACM CSCW (35 papers) (Figure 9). Trends in publication over the years show a significant increase, particularly from 2012 onwards, as illustrated in Figure 10. Notably, the number of papers surged in 2019 (69 papers) and 2020 (54 papers), a period likely influenced by the COVID-19 pandemic, which emphasized the importance of remote collaboration.

The types of user interaction and user experience in collaborative XR environments were also evaluated, categorized as follows: Visual + Haptics + Auditory (189 papers), Visual + Auditory (112 papers), Visual + Haptic (75 papers), and Visual only (22 papers) (see Figure 12b). Author contribution visualization, represented by bubble charts, provides insights of involvement of different authors highlights the level of engagement Figure 11.

To address **SRQ1** on how users effectively connect and collaborate with other users in XR collaboration spaces, the review reveals that effective collaboration in these environments hinges on several key factors. Users connect through various interaction modalities that incorporate visual, haptic, and auditory elements. Technologies such as avatars, spatial annotations, and 360-degree video are integral to facilitating these interactions. Additionally, the design of collaborative interfaces and platforms that support real-time interaction and multi-user engagement is crucial, enhancing the overall effectiveness of collaboration among users in XR settings.

Regarding **SRQ2** on the types of collaboration styles, user experiences, and evaluation techniques used in XR collaboration, the findings show a diverse array of styles ranging from

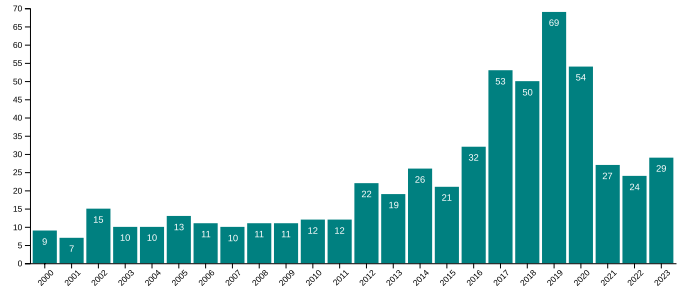


Fig. 10. Distribution of publications by year.

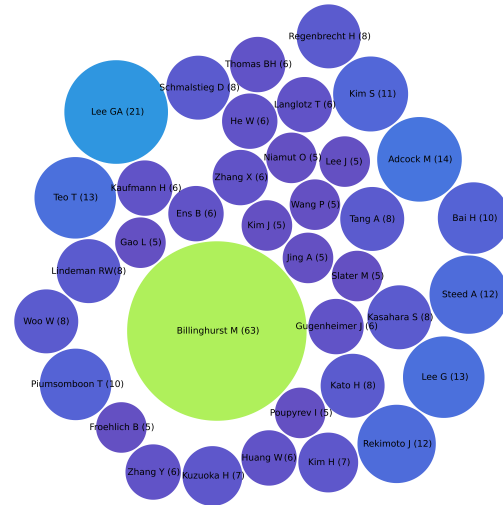


Fig. 11. Author Contribution Visualization: Bubble radius proportional to number of publications.

synchronous to asynchronous interactions, tailored to different tasks and user needs. The user experience is significantly enhanced by immersive technologies that provide realistic sensory feedback, such as haptic cues and spatial audio. Evaluation techniques commonly include user studies, usability testing, and performance metrics to assess the effectiveness of XR applications, focusing on user engagement, satisfaction, and the practical impact of XR technologies on collaborative tasks.

For **SRQ3**, concerning the current trends in XR collaboration and the main research challenges, the review identifies several emerging trends, including the widespread adoption of multi-user platforms, the integration of advanced rendering techniques, and the application of XR in fields like healthcare, education, and remote work. Key research challenges include network optimization, privacy and security concerns, and the need for refined interaction models to better support natural and intuitive user interactions. Addressing these challenges is essential for advancing the capabilities and usability of XR collaboration tools.

The distribution of 395 papers, which were further assessed to determine the current trends in XR collaboration, is organized by year of publication, publication venues, citation impact, type of publication, and area of application, as shown in Figure 4.

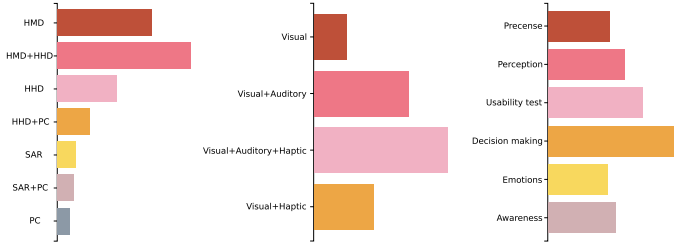


Fig. 12. Device types (a), User interaction types (b), Evaluation criteria (c).

VI. DISCUSSION

We now discuss the results of coding the MR/AR papers focusing on challenges, new trends, and future directions.

A. Challenges

Several significant challenges have been identified in the review. Foremost among these is the absence of safety-certified XR hardware, preventing widespread deployment on a large scale. Furthermore, issues persist with the accuracy and reliability of tracking and mapping technologies, which fall short of ensuring precise assembly and construction processes. Additionally, the limited battery life of XR devices, notably head-mounted displays (HMDs), significantly curtails their usability. These combined obstacles present formidable barriers to the seamless incorporation of AR technologies into collaborative workflows within the XR landscape, underscoring the pressing need for innovative solutions to address these critical limitations. Below we discuss some of the main challenges.

a) Real-time Dynamic 3D mapping of Changing Environments: Maps continually change environments such as construction sites which is essential to capture. The main challenges for 3D mapping construction sites are twofold (1) new components are included constantly and (2) the environment is highly dynamic, workers and machinery are continually moving, materials are transported to different locations. Current mapping technology, such as simultaneous localization and mapping (SLAM) is not sufficiently robust and accurate and has not been tested successfully in such adverse conditions. A mapping system is required that constantly captures and updates the physical world, to enable augmentation and virtualization of dynamic environments.

b) Multi-user Multimodal Capabilities: Multi-user immersive experiences interacting on the same BIM model with multiple viewports can facilitate engagement with clients, designers, and stakeholders. Multiple viewports will help reduce the time to identify anomalies and take effective actions.

c) Merging of Physical and Virtual Worlds: The integration of the physical world into the virtual world and how the virtual world manifests itself in the physical world need to be revised to adapt to collaborative XR systems within which multiple users have their own hardware capabilities and may not be aware of the differences between them. Since more and more XR devices have been marketed to the larger public, it would be necessary to study how mediated environments

created by HMD, smart glasses, projection-based, or CAVE-like systems can be perceived individually by each user and how these differences in display and interaction devices will affect users' roles in the whole collaboration process.

d) Diminished Reality and Real-time Occlusion: Diminished reality uses various technologies to hide real-life objects from the users' view. Real-time occlusion enables realistic augmentations by not rendering parts of virtual objects that are behind physical objects in the perspective of the user's point of view. Both diminished reality and real-time occlusion are necessary capabilities for future AR systems as physical objects are often in the way of augmentations. Diminished reality uses four main techniques: (1) diminish, which degrades or distorts visual fields; (2) see-through, which covers real objects with imagery of their occluded background; (3) replace, which overlaps a virtual object on top of a real object.; and (4) in-paint, which generate plausible background images based on the surrounding environment.

e) Security and Privacy: With the deep integration of AR/MR-based remote collaboration and the internet, users move collaborative tasks to cloud-based systems, therefore security has to be carefully taken into consideration. Whilst many security solutions offered by the current remote collaborative systems are supposed to deal with the transmission level, they can benefit from taking additional security factors into account to protect the remote collaboration data (e.g., the structure of machines, and problem solutions). Data Balancing security and privacy is a complex task in MR development. Striking the right balance ensures that users can safely and confidently engage in MR experiences without undue risks to their data or personal information. Robust security measures, like encryption and secure authentication, protect user data. To facilitate natural interactions within the MR environment, intuitive methods, such as gesture recognition, voice commands, and haptic feedback, are integrated. Data integration with external sources and secure sharing options empower collaboration and data utilization. Quantum communication, such as Quantum Key Distribution (QKD), can be integrated into MR applications to provide an additional layer of security. QKD enables the generation of encryption keys that are theoretically immune to eavesdropping, making communication within the MR environment highly secure. It ensures that confidential information remains protected from even the most advanced cryptographic attacks. Blockchain technology, known for its immutability and transparency, can be utilized for secure data storage and authentication within MR environments. This ensures the integrity of data and transactions. Homomorphic encryption enables computations on encrypted data without revealing the data itself. This is particularly useful for protecting sensitive information while allowing useful computations within MR applications. Secure Multi-Party Computation (SMPC) enables collaborative computation over private inputs, preserving data privacy while supporting cooperative tasks in MR environments.

B. New Trends

Over the previous 23 years, there have been significant updates and changes in XR research trends. Some of the new

research trends and observations were already introduced in the previous section, such as Reconstruction and Collaboration. Here we explicitly present two important changes in XR research that we identified from the 2000–2023 publications such as increases in Evaluation and Rendering research.

a) *Collaborators Size*: Prior work has yet to extensively explore the effects of different sizes of collaborators on collaboration mechanisms, communication dynamics, and social interactions within XR environments, particularly in relation to remote users. Research attention has predominantly been directed towards face-to-face or one-on-one collaboration, largely due to challenges such as network bandwidth constraints and the considerable data transfer demands required for smooth collaboration. Recent research trends concerning collaborative XR systems, focus on how communication and interaction techniques vary across different collaborator sizes, spanning from small dyads to larger groups.

b) *Gen AI*: One emerging trend in XR collaboration research is the integration of Artificial Intelligence (AI), Machine Learning (ML), and Generative AI (GenAI). These technologies enhance collaborative experiences by analyzing user behaviors, personalizing interactions, and generating realistic virtual environments. AI-driven interfaces and assistants utilize NLP, computer vision, and gesture recognition to facilitate seamless interactions between users and the virtual environment. Machine learning algorithms, including Generative AI models, optimize XR systems based on user feedback, dynamically adjusting parameters and content recommendations.

c) *Digital Twin (DT)*: A digital twin serves as a virtual counterpart to a physical system or object, intimately linked to it through sensors and data collection devices. Digital twins are set to play a pivotal role in inducing the sense of presence and facilitating remote assistance by providing collaborators with a virtual representation of physical systems, bridging the gap between remote and local participants [52], [53]. This fosters a shared understanding of the collaborative environment and enables seamless remote assistance and support. Especially in healthcare, digital twins enable remote assistance and diagnosis, allowing experts to collaborate with frontline workers in real-time, thereby improving patient care and outcomes [54].

d) *Avatars/Personas*: The concept of human-like avatars, or personas, emerges as a transformative force in XR collaboration. These personas, with their lifelike appearance and behavior, redefine the way users interact and collaborate in virtual environments. As technology progresses and the uncanny valley phenomenon is gradually mitigated, these human-like avatars are becoming increasingly realistic and capable of facilitating more natural and immersive interactions. They evoke a stronger sense of social presence, fostering a deeper connection between users in virtual environments also increased engagement and empathy.

e) *Spatial Computing*: Spatial computing technologies enable XR devices to map and understand physical spaces, allowing users to interact with virtual content in real-world environments. A crucial component of spatial computing is the capability of six degrees of freedom (6DoF), which empowers users to move freely and manipulate objects in three-dimensional space [55]. By leveraging 6DoF, XR collabora-

tion platforms offer enhanced spatial interaction, immersive experiences, and seamless collaboration, revolutionizing the way teams work together across geographical distances and timezones.

C. Future Direction

In future directions, advancing XR collaboration systems will likely focus on several key areas.

a) *Real-time Environments Mapping*: The need for real-time 3D mapping of constantly changing environments, such as those encountered in construction sites. The challenges faced in XR collaboration related to this include:

Handling Dynamic Environments: XR collaboration scenarios often involve real-time interaction in dynamic spaces where the environment is subject to continuous changes. This could include collaborative work in construction, remote inspection of changing manufacturing processes, or collaborative exploration of evolving landscapes.

Data Accuracy and Robustness: XR collaboration relies on precise spatial mapping and tracking of objects and users. Ensuring the accuracy and robustness of 3D mapping technologies in these dynamic environments is a significant challenge. Inaccurate or outdated mapping data can lead to misalignments and disruptions in collaborative XR experiences.

Integration of Emerging Technologies: To address these challenges, there is a growing need to integrate emerging technologies, such as drones equipped with advanced sensors and cameras, into XR collaboration setups. These technologies can facilitate real-time data capture and mapping updates in dynamic environments.

b) *Tech*: The advancements in mobile connectivity, CPU and GPU processing capabilities, wearable form factors, tracking reliability, and the accessibility of commercial-grade game creation tools are driving increased interest in using XR technologies for entertainment purposes. Both AR and VR research demonstrate the immersive potential of these technologies, offering users a deeper sense of presence and opening up new avenues for enjoyment within MR environments. For entertainment-focused AR applications, prioritizing natural user interaction is crucial to ensuring sustained usage, as users may be deterred by technologies that induce fatigue over time. Therefore, wearable AR devices are likely to be more appealing for entertainment XR applications. Exploring collaborative XR interaction concepts, such as shared cues, gaze-based interaction, eye contact, handshakes, and other non-verbal interactions, warrants further investigation. This holistic approach can enrich the XR collaboration experience, promoting innovation and enhancing user engagement across diverse application domains.

c) *Integration of AI and IoT*: Integration with Artificial Intelligence (AI) and Internet of Things (IoT) technologies will enhance XR collaboration systems, enabling intelligent features such as context-aware interactions, real-time data analysis, and predictive modeling to support decision-making and task completion.

d) *Multi-user Multimodal Capabilities*: The need for XR collaboration platforms to support multiple users interacting

with shared models or environments using a variety of input methods and modes of interaction is crucial. This introduces the challenge of ensuring seamless synchronization and communication among users engaged in collaborative tasks. XR collaboration systems must address the complexities of managing user interactions and data exchange while maintaining real-time synchronization and preserving the integrity of the shared XR space. The challenge is to design and implement XR collaboration solutions that can handle diverse user inputs and interaction modes efficiently, allowing for effective communication and collaboration among participants. Additionally, ensuring a consistent and engaging user experience across different devices and modalities can be a significant challenge in this context.

e) Merging Spaces: The need for XR systems to seamlessly integrate both physical and virtual environments is important for collaboration. The challenge lies in ensuring that users with diverse XR hardware capabilities can collaborate effectively, even when they may not be fully aware of the differences in their hardware setups. With the increasing availability of various XR devices to the general public, it becomes important to investigate how each user perceives the mediated environments created by different hardware types, such as head-mounted displays, smart glasses, projection based systems, or CAVE-like environments. This challenge involves understanding how these differences in display and interaction devices can impact users' roles and experiences in collaborative XR settings, ultimately striving for a cohesive and inclusive collaborative environment.

VII. CONCLUSION

In this systematic review, we have analyzed 557 papers that reported various form of collaboration in the field of XR. The review has confirmed that two key aspects are central to XR collaboration. Technology centric evaluations involve benchmarking aspects such as tracking, displays, reconstruction, rendering, and calibration. Human-centric evaluations encompass the assessment of applications, design, and human factors. Additionally, the review identified a limited number of team-centric evaluations that involve collaboration, communication, and understanding of environments and work practices. Furthermore, the importance of multi-user capabilities has been highlighted in previous research, along with the need for a more diverse and inclusive gender-balanced participant population in user studies.

There is a growing trend towards integrating various XR technologies to create more immersive and collaborative environments. With the growing influence of remote work and distributed teams, XR systems are increasingly being used to facilitate collaboration. Efforts are being made to develop XR collaboration platforms that are compatible across different devices and settings. User experience research has focused on improving collaboration environments through features such as natural hand gestures, spatial audio, and realistic avatars. We also identified XR collaboration being adopted and applied across various industries, including healthcare, education, manufacturing, and entertainment, each with its own unique applications and requirements.

We think this review will be useful to researchers and students in academia and industry practitioners as it offers a comprehensive summary of current research trends in Collaborative XR and highlights intriguing topics for further opportunities. Overall, we hope this research will help XR integrate seamlessly into our daily lives and positively influence people in a variety of ways, such as how we think, feel, behave, or communicate in the future.

REFERENCES

- [1] P. Arunprasad, C. Dey, F. Jebli, A. Manimuthu, and Z. El Hathat, "Exploring the remote work challenges in the era of covid-19 pandemic: review and application model," *Benchmarking: An International Journal*, vol. 29, no. 10, pp. 3333–3355, 2022.
- [2] L. Yang, D. Holtz, S. Jaffe, S. Suri, S. Sinha, J. Weston, C. Joyce, N. Shah, K. Sherman, B. Hecht *et al.*, "The effects of remote work on collaboration among information workers," *Nature human behaviour*, vol. 6, no. 1, pp. 43–54, 2022.
- [3] P. A. Rauschnabel, R. Felix, C. Hinsch, H. Shahab, and F. Alt, "What is xr? towards a framework for augmented and virtual reality," *Computers in human behavior*, vol. 133, p. 107289, 2022.
- [4] C. G. Fidalgo, Y. Yan, H. Cho, M. Sousa, D. Lindlbauer, and J. Jorge, "A survey on remote assistance and training in mixed reality environments," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 5, pp. 2291–2303, 2023.
- [5] S. Patel, B. Panchotiya, A. Patel, A. Budharani, and S. Ribadiya, "A survey: virtual, augmented and mixed reality in education," *International Journal of Engineering Research & Technology (IJERT)*, vol. 9, no. 05, p. 2020, 2020.
- [6] J. A. De Guzman, K. Thilakarathna, and A. Seneviratne, "Security and privacy approaches in mixed reality: A literature survey," *ACM Computing Surveys (CSUR)*, vol. 52, no. 6, pp. 1–37, 2019.
- [7] M. K. Bekele, R. Pierdicca, E. Frontoni, E. S. Malinvernì, and J. Gain, "A survey of augmented, virtual, and mixed reality for cultural heritage," *Journal on Computing and Cultural Heritage (JOCCH)*, vol. 11, no. 2, pp. 1–36, 2018.
- [8] X. Castañer and N. Oliveira, "Collaboration, coordination, and cooperation among organizations: Establishing the distinctive meanings of these terms through a systematic literature review," *Journal of management*, vol. 46, no. 6, pp. 965–1001, 2020.
- [9] C. Ellis and J. Wainer, "Groupware and computer supported cooperative work," *Multiagent systems: a modern approach to distributed artificial intelligence*, pp. 425–457, 1999.
- [10] S. Greenberg and D. Marwood, "Real time groupware as a distributed system: concurrency control and its effect on the interface," in *Proceedings of the ACM Conference on Computer Supported Cooperative Work*, ser. CSCW '94. ACM, 1994, p. 207–217. [Online]. Available: <https://doi.org/10.1145/192844.193011>
- [11] R. M. Baecker, J. Grudin, W. A. Buxton, and S. Greenberg, "Chapter 11 - groupware and computer-supported cooperative work," in *Readings in Human-Computer Interaction*, ser. Interactive Technologies, R. M. Baecker, J. Grudin, W. A. Buxton, and S. Greenberg, Eds. Morgan Kaufmann, 1995, pp. 741–753. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/B9780080515748500777>
- [12] L. M. Applegate, "Technology support for cooperative work: A framework for studying introduction and assimilation in organizations," *Journal of Organizational Computing*, vol. 1, no. 1, pp. 11–39, 1991. [Online]. Available: <https://doi.org/10.1080/10919399109540148>
- [13] S. Mann, T. Furness, Y. Yuan, J. Iorio, and Z. Wang, "All reality: Virtual, augmented, mixed (x), mediated (x, y), and multim mediated reality," *arXiv preprint arXiv:1804.08386*, 2018.
- [14] P. Milgram and F. Kishino, "A taxonomy of mixed reality visual displays," *IEICE TRANSACTIONS on Information and Systems*, vol. 77, no. 12, pp. 1321–1329, 1994.
- [15] M. Al-Adhami, S. Wu, and L. Ma, "Extended reality approach for construction quality control," in *Extended reality approach for construction quality control*. International Council for Research and Innovation in Building and ..., 2019.
- [16] J. A. Paradiso and J. A. Landay, "Guest editors' introduction: Cross-reality environments," *IEEE Pervasive Computing*, vol. 8, no. 3, pp. 14–15, 2009.

- [17] J. R. Wallace, S. Oji, and C. Anslow, "Technologies, methods, and values: Changes in empirical research at cscw 1990-2015," *Proceedings of the ACM on Human-Computer Interaction*, vol. 1, no. CSCW, pp. 1–18, 2017.
- [18] C. Boletsis, "The new era of virtual reality locomotion: A systematic literature review of techniques and a proposed typology," *Multimodal Technologies and Interaction*, vol. 1, no. 4, p. 24, 2017.
- [19] A. Dey, M. Billinghurst, R. W. Lindeman, and J. Swan, "A systematic review of 10 years of augmented reality usability studies: 2005 to 2014," *Frontiers in Robotics and AI*, vol. 5, p. 37, 2018.
- [20] R. A. J. de Belen, H. Nguyen, D. Filonik, D. Del Favero, and T. Bednarz, "A systematic review of the current state of collaborative mixed reality technologies: 2013–2018," *AIMS Electronics and Electrical Engineering*, vol. 3, no. 2, p. 181, 2019.
- [21] B. Ens, J. Lanir, A. Tang, S. Bateman, G. Lee, T. Piumsomboon, and M. Billinghurst, "Revisiting collaboration through mixed reality: The evolution of groupware," *International Journal of Human-Computer Studies*, vol. 131, pp. 81–98, 2019.
- [22] L. Merino, M. Schwarzl, M. Kraus, M. Sedlmair, D. Schmalstieg, and D. Weiskopf, "Evaluating mixed and augmented reality: A systematic literature review (2009-2019)," *arXiv preprint arXiv:2010.05988*, 2020.
- [23] M. Adcock and C. Gunn, "Using projected light for mobile remote guidance," *Computer Supported Cooperative Work (CSCW)*, vol. 24, no. 6, pp. 591–611, 2015.
- [24] B. Jones, A. Wicraft, S. Bateman, C. Neustaedter, and A. Tang, "Mechanics of camera work in mobile video collaboration," in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2015, pp. 957–966.
- [25] P. Wang, S. Zhang, X. Bai, M. Billinghurst, W. He, M. Sun, Y. Chen, H. Lv, and H. Ji, "2.5 dhands: a gesture-based mr remote collaborative platform," *The International Journal of Advanced Manufacturing Technology*, vol. 102, no. 5, pp. 1339–1353, 2019.
- [26] D. Cao and S. Kim, "Immediate presence of handwritten annotation on moving picture in real-time collaborative system," *IEEE Access*, vol. 8, pp. 133 258–133 268, 2020.
- [27] S. Gauglitz, B. Nuernberger, M. Turk, and T. Höllerer, "In touch with the remote world: Remote collaboration with augmented reality drawings and virtual navigation," in *Proceedings of the 20th ACM Symposium on Virtual Reality Software and Technology*, 2014, pp. 197–205.
- [28] J. Pejosa-Laaajola, S. Reponen, M. Virnes, and T. Leinonen, "Mobile augmented communication for remote collaboration in a physical work context," *Australasian Journal of Educational Technology*, vol. 33, no. 6, 2017.
- [29] P. Mohr, S. Mori, T. Langlotz, B. H. Thomas, D. Schmalstieg, and D. Kalkofen, "Mixed reality light fields for interactive remote assistance," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020, pp. 1–12.
- [30] T. Teo, G. A. Lee, M. Billinghurst, and M. Adcock, "360drops: Mixed reality remote collaboration using 360 panoramas within the 3d scene," in *SIGGRAPH Asia 2019 Emerging Technologies*. SIGGRAPH, 2019, pp. 1–2.
- [31] T. Rhee, S. Thompson, D. Medeiros, R. dos Anjos, and A. Chalmers, "Augmented virtual teleportation for high-fidelity telecollaboration," *IEEE Transactions on Visualization and Computer Graphics*, vol. 26, no. 5, pp. 1923–1933, 2020.
- [32] M. Le Chénéchal, T. Duval, V. Gouranton, J. Royan, and B. Arnaldi, "Vishnu: virtual immersive support for helping users an interaction paradigm for collaborative remote guiding in mixed reality," in *2016 IEEE Third VR International Workshop on Collaborative Virtual Environments (3DCVE)*. IEEE, 2016, pp. 9–12.
- [33] L. Gao, H. Bai, G. Lee, and M. Billinghurst, "An oriented point-cloud view for mr remote collaboration," in *SIGGRAPH ASIA 2016 Mobile Graphics and Interactive Applications*. , 2016, pp. 1–4.
- [34] J. Venerella, L. Sherpa, H. Tang, and Z. Zhui, "A lightweight mobile remote collaboration using mixed reality," in *Proceedings of Computer Vision and Pattern Recognition*, 2019, pp. 1–4.
- [35] D. Reilly, M. Salimian, B. MacKay, N. Mathiasen, W. K. Edwards, and J. Franz, "Secspace: prototyping usable privacy and security for mixed reality collaborative environments," in *Proceedings of the 2014 ACM SIGCHI symposium on Engineering interactive computing systems*, 2014, pp. 273–282.
- [36] J. Reinhardt and K. Wolf, "Go-through: Disabling collision to access obstructed paths and open occluded views in social vr," in *Proceedings of the Augmented Humans International Conference*, 2020, pp. 1–10.
- [37] M. Niederhauser, W. Allsbrook, E. Zananiri, and J. Fitzgerald, "Metamorphic: A social vr experience," in *ACM SIGGRAPH 2020 Immersive Pavilion*. ACM SIGGRAPH, 2020, pp. 1–2.
- [38] P. Wang, X. Bai, M. Billinghurst, S. Zhang, S. Wei, G. Xu, W. He, X. Zhang, and J. Zhang, "3dgam: using 3d gesture and cad models for training on mixed reality remote collaboration," *Multimedia Tools and Applications*, pp. 1–26, 2020.
- [39] C. Skourup, T. G. Graven, and K. Husøy, "Can visualisation improve remote collaboration?" *IFAC Proceedings Volumes*, vol. 40, no. 16, pp. 251–256, 2007.
- [40] J. Lee, M. Kim, and J. Kim, "Rolevr: Multi-experience in immersive virtual reality between co-located hmd and non-hmd users," *Multimedia Tools and Applications*, vol. 79, no. 1, pp. 979–1005, 2020.
- [41] M. Kim and K. Wohn, "Holobox: Augmented visualization and presentation with spatially integrated presenter," *Interacting with Computers*, vol. 30, no. 3, pp. 224–242, 2018.
- [42] H. Kato, M. Billinghurst, I. Poupyrev, K. Imamoto, and K. Tachibana, "Virtual object manipulation on a table-top ar environment," in *Proceedings IEEE and ACM International Symposium on Augmented Reality (ISAR 2000)*. Ieee, 2000, pp. 111–119.
- [43] M. Faridan, B. Kumari, and R. Suzuki, "Chameleoncontrol: Teleoperating real human surrogates through mixed reality gestural guidance for remote hands-on classrooms," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–13.
- [44] G. Goebbels and V. Lalioti, "Co-presence and co-working in distributed collaborative virtual environments," in *Proceedings of the 1st international conference on Computer graphics, virtual reality and visualisation*, 2001, pp. 109–114.
- [45] F. Welsford-Ackroyd, A. Chalmers, R. Kuffner dos Anjos, D. Medeiros, H. Kim, and T. Rhee, "Spectator view: Enabling asymmetric interaction between hmd wearers and spectators with a large display," *Proceedings of the ACM on Human-Computer Interaction*, vol. 5, no. ISS, pp. 1–17, 2021.
- [46] T. Pejosa, J. Kantor, H. Benko, E. Ofek, and A. Wilson, "Room2room: Enabling life-size telepresence in a projected augmented reality environment," in *Proceedings of the 19th ACM conference on computer-supported cooperative work & social computing*, 2016, pp. 1716–1725.
- [47] G. Reitmayr and D. Schmalstieg, "Mobile collaborative augmented reality," in *Proceedings IEEE and ACM International Symposium on Augmented Reality*. IEEE, 2001, pp. 114–123.
- [48] J. Müller, R. Rädle, and H. Reiterer, "Remote collaboration with mixed reality displays: How shared virtual landmarks facilitate spatial referencing," in *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 2017, pp. 6481–6486.
- [49] J. G. Grandi, I. Berndt, H. G. Debarba, L. Nedel, and A. Maciel, "Collaborative manipulation of 3d virtual objects in augmented reality scenarios using mobile devices," in *2017 IEEE Symposium on 3D User Interfaces (3DUI)*. IEEE, 2017, pp. 264–265.
- [50] K. Jeong, J. Kim, M. Kim, J. Lee, and C. Kim, "Asymmetric interface: user interface of asymmetric virtual reality for new presence and experience," *Symmetry*, vol. 12, no. 1, p. 53, 2020.
- [51] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent dirichlet allocation," *Journal of machine Learning research*, vol. 3, no. Jan, pp. 993–1022, 2003.
- [52] Q. Qi, F. Tao, Y. Zuo, and D. Zhao, "Digital twin service towards smart manufacturing," *Procedia Cirp*, vol. 72, pp. 237–242, 2018.
- [53] A. A. Malik and A. Bilberg, "Digital twins of human robot collaboration in a production setting," *Procedia manufacturing*, vol. 17, pp. 278–285, 2018.
- [54] F. Tao, Q. Qi, L. Wang, and A. Nee, "Digital twins and cyber-physical systems toward smart manufacturing and industry 4.0: Correlation and comparison," *Engineering*, vol. 5, no. 4, pp. 653–661, 2019.
- [55] R. Chen, F.-L. Zhang, S. Finnie, A. Chalmers, and T. Rhee, "Casual 6-dof: free-viewpoint panorama using a handheld 360 camera," *IEEE Transactions on Visualization and Computer Graphics*, 2022.